



The Database Demands of Peta-scale Computing

In concluding this two-part series, we take a brief look at the challenges for next-generation databases.

Part—2

In the previous article we discussed that as data volumes grow towards the peta-scale and beyond, most traditional databases find it difficult to scale up to the needs of cloud computing and big data. While big data drives the need for databases to handle huge volumes of data, cloud computing requires dynamic scalability.

Dynamic scalability

Dynamic scalability, at first glance, appears simple to achieve in databases with the Shared Nothing architecture. This architecture splits the database into multiple partitions, with one partition per server. Therefore, it should be possible to dynamically scale by adding another server and repartitioning the database across all servers. However, database partitioning is an important factor in the performance of database access. Database queries frequently access related sets of information. If a frequently accessed related set of data gets partitioned across multiple servers, data from multiple partitions needs to be accessed before the query results can be returned—a process known as data shipping. This can have a large adverse impact on query access times. Data partitioning needs to take data access patterns into account in order to reduce data shipping costs. Hence, 'Shared Nothing' traditional relational databases typically find it difficult to achieve dynamic scalability.

The limitation of the traditional RDBMS in being able to scale dynamically led to the NoSQL movement, and the development of special-purpose data-stores (since the term database is typically used to represent an ACID-compliant traditional database, we use the term data-store for the NoSQL databases). They proposed data persistent stores typically built as replicated data tables, which trade off strict ACID semantics for improved scalability. These systems are characterised by eventual consistency (which we discussed in last month's article) instead of the strict ACID consistency of RDBMSs. While NoSQL databases such as BigTable, Cassandra or SimpleDB have shown excellent performance, it is not yet clear whether they can scale to the demands of the next decade's peta-scale computing.

Big data challenges

Over the past few decades, there have been two major camps of database technology. The first one is that of OnLine Transaction Processing (OLTP) workloads, and the second that of OnLine Analytical Processing (OLAP) workloads. An OLTP workload typically consists of a number of concurrent transactions that perform insertions, deletions and updates of the data stored in the database. OLTP workloads are both read- and write-intensive. Popular examples that we encounter in everyday life are bank